August 28, 2026 10:00 AM GMT

US Rates Strategy | North America

A 40 Trillion Dollar Distraction

Does debt create problems? If so, figuring out how much of it investors will buy hasn't been one. Deficits – which add to the debt – may create too much inflation, but the real problem with debt involves predicting the price at which it trades. $40 trillion is just a number, and a distraction.

Key Takeaways

- In October 2022, the US public debt hit $31 trillion and we wrote "The 31 Trillion Dollar Distraction" in response to the media coverage that resulted.
- Since then, the US public debt has climbed to $40 trillion and the media responded with more warnings about the fiscal largesse of the federal government.
- In October 2022, 10y yields reached 4.35% – 40bp lower than their recent 4.75% peak – putting into perspective the concession for the additional $9 trillion.
- We think the CBO's 4.4% long-term projection for 10y Treasury yields overstates them by at least 100bp, if core PCE inflation returns to 2% as it projects.
- In what follows, we embellish our October 2022 note with additional perspective on the public debt and purposefully abstain from any fear-mongering.

Click here for our Bloomberg Ticker Almanac.

Please add me to your distribution list.

MORGAN STANLEY & CO. LLC

Matthew Hornbach
Strategist
Matthew.Hornbach@morganstanley.com +1 212 761-1837

Martin W Tobias, CFA
Strategist
Martin.Tobias@morganstanley.com +1 212 761-6076

Shaun Zhou
Strategist
Shaun.Zhou@morganstanley.com +1 212 761-3348

Aryaman Singh
Strategist
Aryaman@morganstanley.com +1 212 761-1993

Eli P Carter
Strategist
Eli.Carter@morganstanley.com +1 212 761-4703

Interest Rate Strategy

United States | Keep calm and carry on

MORGAN STANLEY & CO. LLC

Matthew Hornbach, CMT
Matthew.Hornbach@morganstanley.com +1 212 761-1863

Martin Tobias, CFA, CMT
Martin.Tobias@morganstanley.com +1 212 761-6076

Shaun Zhou
Shaun.Zhou@morganstanley.com +1 212 761-6076

Aryaman Singh
Aryaman@morganstanley.com +1 212 761-1993

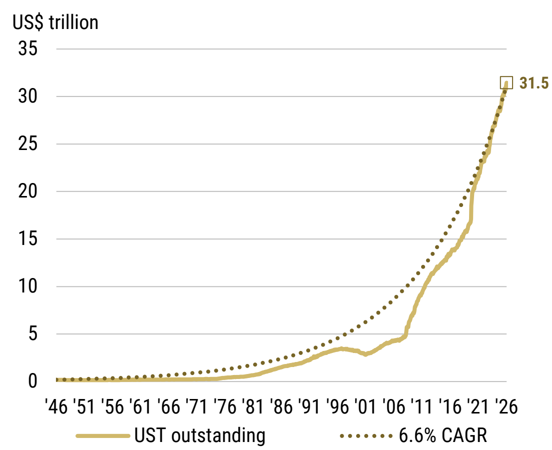
Eli Carter
Eli.Carter@morganstanley.com +1 212 761-4703

The debt has compounded at 6.6% since 1946

US public debt made breaking news headlines by rising above $40 trillion, and the coverage repeated the warnings that followed the move above $31 trillion in October 2022. Ten-year Treasury yields recently peaked at 4.75%, or 40bp above the 4.35% level they reached in October 2022, which puts the concession investors demanded for the additional $9 trillion into perspective.

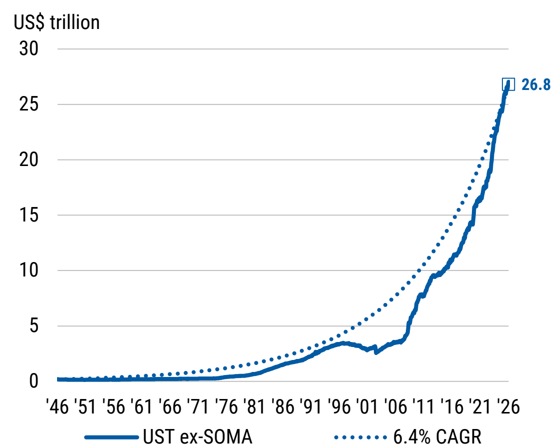
The rate at which the debt accumulates has barely changed over the past decade. Marketable Treasury debt outstanding has compounded at 6.6% a year since 1946, and at 6.4% a year excluding the securities held in the Federal Reserve's System Open Market Account (SOMA), as shown in Exhibit 1 and Exhibit 2 .

Exhibit 1: UST outstanding and 6.6% compound annual growth rate



Source: Morgan Stanley Research, US Treasury

Exhibit 2: UST outstanding ex-SOMA and 6.4% compound annual growth rate



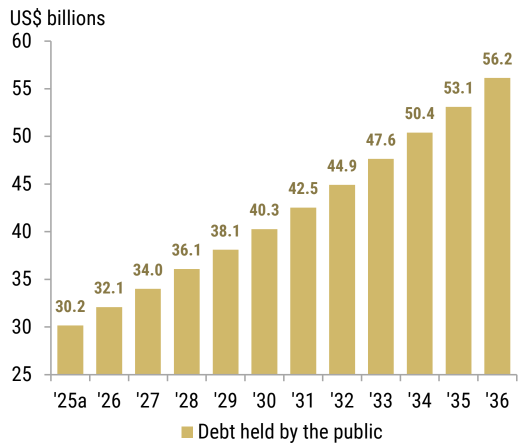
Source: Morgan Stanley Research, US Treasury

If $40 trillion presents a problem today, then the same problem presented itself at $1 trillion in the early 1980s and at $10 trillion in 2008. Concern about the level of the debt has recurred at each round number without a visible change in the growth rate of the debt, save for the austerity in the late 1990s to early noughties.

The forward path draws most of the attention. The Congressional Budget Office (CBO) projects federal debt held by the public to rise from about $31 trillion today to $47 trillion in 2033 and $56 trillion in 2036 (see Exhibit 3).

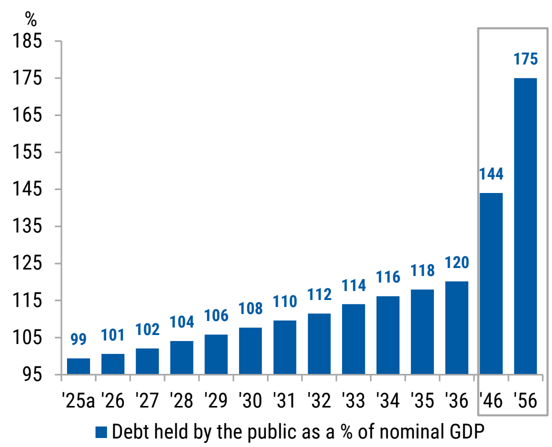
- By the time college graduates today are ready to retire, the CBO projects US federal debt at 175% of GDP, having passed the 106% record set in 1946 in 2030 (see Exhibit 4).

Exhibit 3: CBO baseline projection of federal debt held by the public



Source: Morgan Stanley Research, Congressional Budget Office

Exhibit 4: CBO extended baseline projection of federal debt held by the public as % of GDP



Source: Morgan Stanley Research, Congressional Budget Office

Before comparing any of these numbers, investors should know the measure they use. Total public debt outstanding, at roughly $40 trillion, includes nonmarketable securities, most of which sit in government trust funds. Marketable Treasury debt outstanding runs closer to $31.5 trillion and marketable debt outstanding ex-SOMA closer to $26.8 trillion.

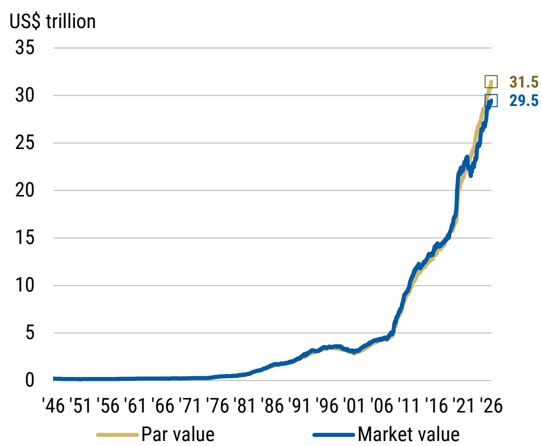
Those three measures differ by more than $13 trillion. Changing the measure moves the reported size of the debt by more than the entire increase that generated the recent media coverage.

Whether a series reports par value or market value matters as well. In normal times the difference distracts, but in times of abnormal volatility the difference is meaningful (see Exhibit 5 and Exhibit 6). Treasury holdings data are mostly reported at market value, while primary dealer transaction data that we reference below are reported at par value.

The size of the debt distracts from what matters most about it:

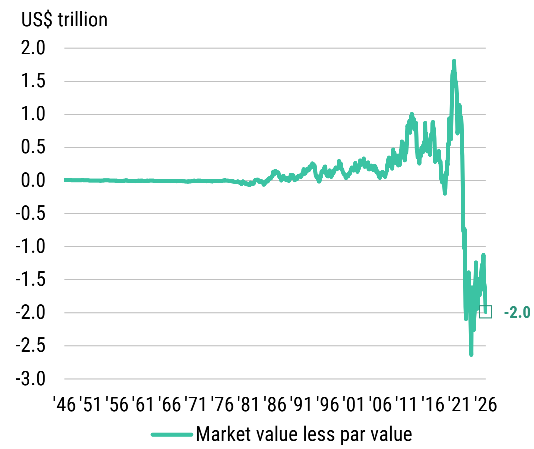
- The speed at which the debt accumulates, i.e., the size and timing of fiscal deficits,
- The risk characteristics of the debt that investors will buy,
- The price at which investors will buy and the value it provides at that price, and
- The major drivers of the yields in the marketplace for it.

Exhibit 5: US Treasury marketable debt outstanding: market value and par value



Source: Morgan Stanley Research, FRB Dallas, Bloomberg

Exhibit 6: US Treasury marketable debt outstanding: market value less par value



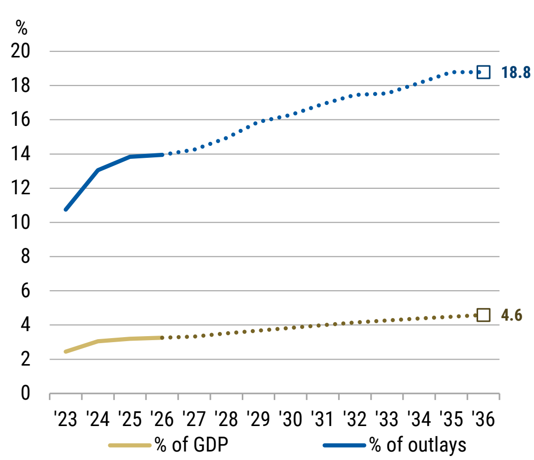
Source: Morgan Stanley Research, FRB Dallas, Bloomberg

Interest-cost projections depend on rate forecasts

Concerns around US debt usually center on interest payments. The CBO projects net interest outlays to rise from 3.3% of GDP in fiscal 2026 to 4.6% in 2036 (and 6.9% by 2056 on its extended baseline), and to grow from about 14% of all federal spending to roughly 19% over the next decade (see Exhibit 7).

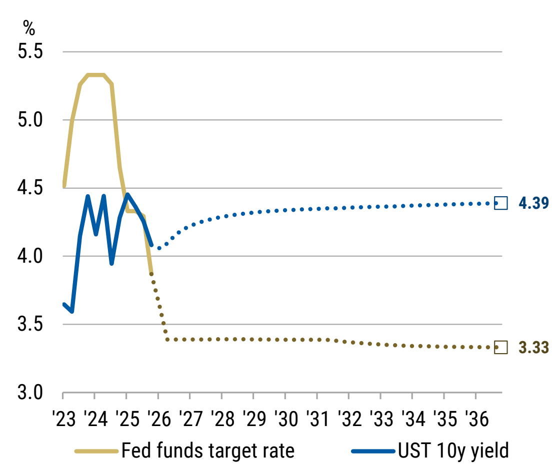
Those projections rest on an interest rate forecast. The CBO's February 2026 baseline assumes a 10y Treasury yield of 4.1% in 2026, rising to 4.4% by 2031 and holding near that level thereafter (see Exhibit 8). Comparing that path against the projected path for Fed policy suggests the Fed cuts rates 50bp from current levels with the 10y Treasury yield falling about 25bp – resulting in a 2s10s curve near 100bp.

Exhibit 7: CBO baseline projections for net interest, as % of total outlays and % of GDP



Source: Morgan Stanley Research, Congressional Budget Office

Exhibit 8: CBO baseline projection for UST 10y yield and the fed funds rate



Source: Morgan Stanley Research, Congressional Budget Office

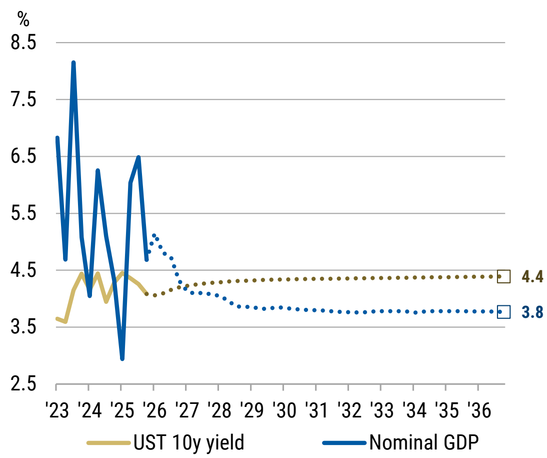
The CBO argues that high and rising debt could increase the likelihood of a fiscal crisis because investors might lose confidence in the US government's ability to service and repay its debt. The CBO also argues that it could lead to higher inflation expectations, erode confidence in the US dollar as a reserve currency, and constrain policymakers from using deficits in a countercyclical way.

We liken these concerns – which the media only seem highlight when 10y yields near periodic highs – to an adult not being able to get over the common childhood fear of a bogeyman hiding under the bed, even though the bogeyman hasn't appeared, ever.

In addition, we can't find a precedent in the past 75 years for the 10y Treasury yield trading >50bp above nominal GDP growth for any length of time, as the CBO projects it to (see Exhibit 9).

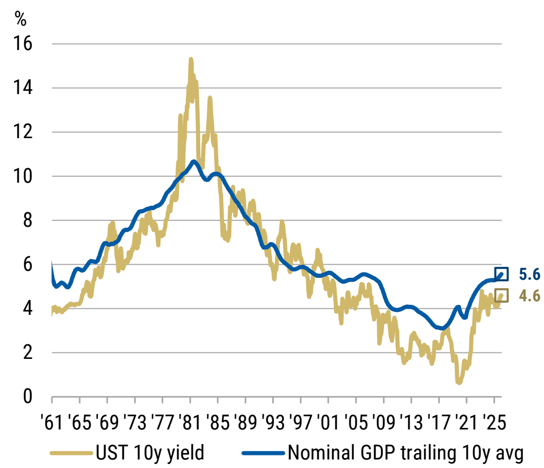
- In fact, since the 1960s, the 10y yield averaged 60bp *below* the trailing 10-year average of nominal GDP - suggesting a more realistic projection for 10y yields at 3.2% instead of 4.4% (see Exhibit 10).

Exhibit 9: CBO baseline projection for UST 10y yield and nominal GDP growth Y/Y



Source: Morgan Stanley Research, Congressional Budget Office

Exhibit 10: UST 10y yield and trailing 10-year average of nominal GDP growth Y/Y

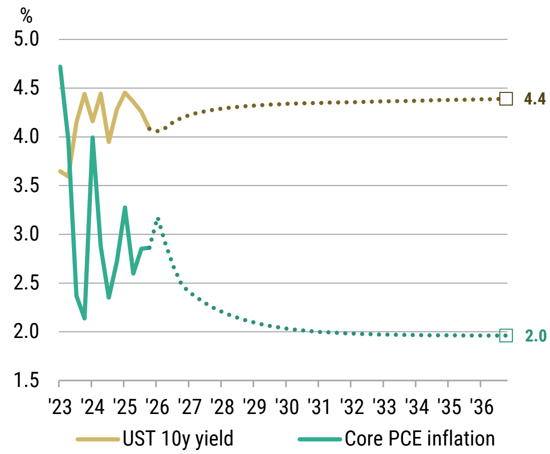


Source: Morgan Stanley Research, BEA, Bloomberg

Furthermore, if we consider the CBO's projection for core PCE inflation to return to 2% by 2030, we see that its forecast for 10y yields looks even more overstated (see Exhibit 11).

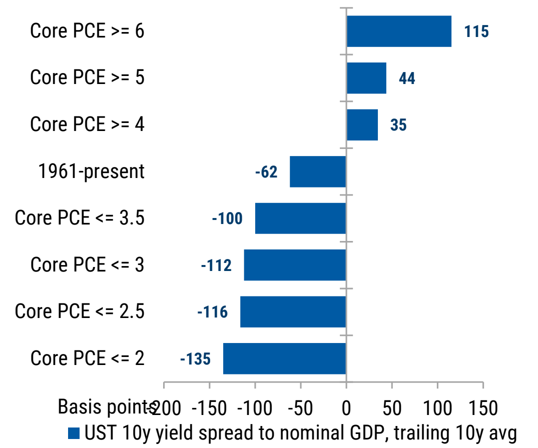
- Conditioning the spread between 10y yields and the trailing 10-year average of nominal GDP on core PCE inflation, we could project 10y yields between 110 and 135bp below the CBO's 3.8% nominal GDP projection, i.e., 2.45-70%.

Exhibit 11: CBO baseline projection for UST 10y yield and core PCE price inflation Y/Y



Source: Morgan Stanley Research, Congressional Budget Office

Exhibit 12: UST 10y yield spread to trailing 10y average of nominal GDP growth, conditional on core PCE inflation



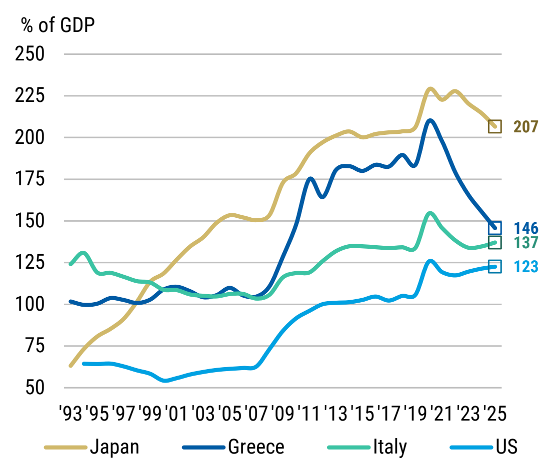
Source: Morgan Stanley Research, BEA, Bloomberg

Yields descend as debt-to-GDP ratios ascend

Government bond yields cannot be linked to the metrics most often cited in the analysis of debt sustainability. The yields on 30-year government bonds in the four G10 countries with the largest general government gross debt-to-GDP ratios, Japan, Greece, Italy, and the US, descend as debt-to-GDP ascends (see Exhibit 13 and Exhibit 14).

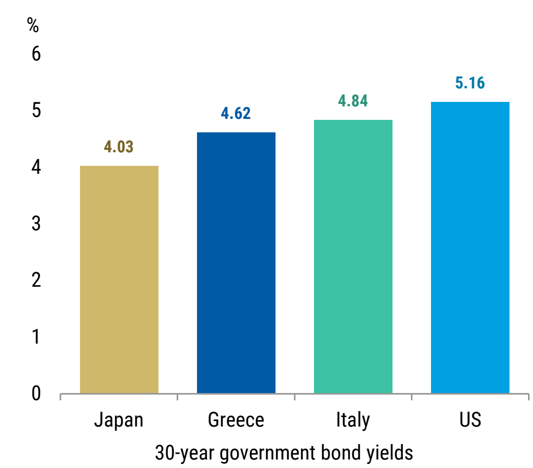
The government debt load in Japan has stood as a notable counterpoint to concerns of this nature for decades. With gross general government debt above 200% of GDP, Japan's situation mitigates some of the CBO's trepidations.

Exhibit 13: General government gross debt-to-GDP: Japan, Greece, Italy, and the US



Source: Morgan Stanley Research, IMF, US Treasury, Bloomberg

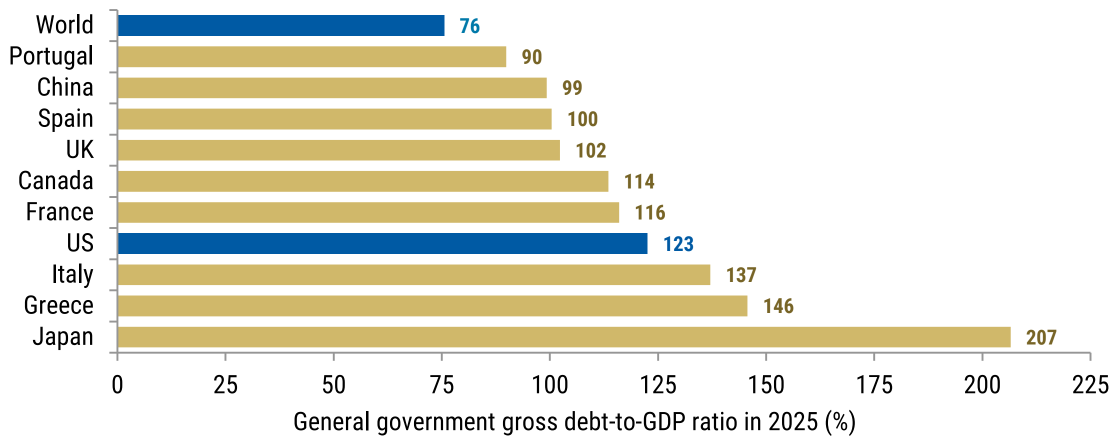
Exhibit 14: 30-year government bond yields: Japan, Greece, Italy, and the US



Source: Morgan Stanley Research, Bloomberg

More generally, debt sustainability concerns tend to suffer from myopia. Investors usually apply a broader perspective than individual worksheets in Excel, and the US hardly stands on its own debt-to-GDP pedestal. Plenty of other countries, with monetary systems similar to the US, operate unfazed with debt-to-GDP ratios above the global average of 76%, as of 2025 (see Exhibit 15).

Exhibit 15: General government gross debt-to-GDP ratio by country



Source: Morgan Stanley Research, IMF, US Treasury, Bloomberg

Currency denomination changes the risk profile

In our view, debt sustainability concerns, which occasionally lead to dramatic but temporary moves in related bond markets, miss important context relating to three things: the structure of the monetary system, the broader world of non-sovereign investment alternatives, and investor psychology.

Debt issued by a government in control of its own currency, e.g., the US and Japan, surely differs in risk profile from debt issued by a government not in control of its own currency, e.g., Greece and Italy, and for many years before the European sovereign debt crisis, emerging market (EM) debt.

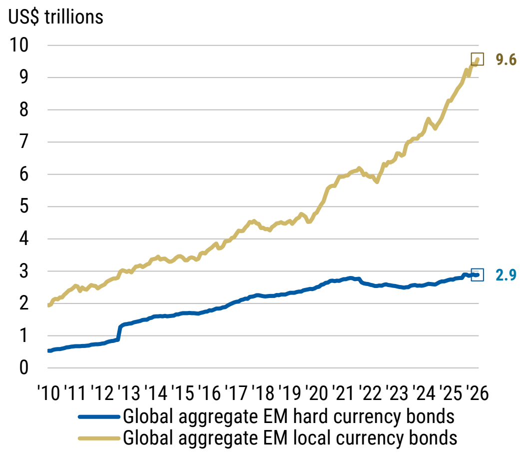
Modern monetary theory (should have) laid to rest concerns about government debt in countries that issue in local currency, and while some exceptions exist, such as a euro area member that can issue in euros but cannot print them by fiat, the US is not one.

Hard-currency debt, whether issued by an EM government or by a sovereign in the euro area, risks debt sustainability backlash much more so than local currency debt, meaning debt issued in the same currency as the sovereign controls, e.g., the US, Canada, Australia, New Zealand, and Japan.

Exhibit 16 and Exhibit 17 show the evolution of EM hard-currency and local-currency bonds outstanding since the Great Financial Crisis (GFC). After the Greek debt crisis in 2012, EM countries issued more hard-currency debt than local-currency debt, increasing exposure to possible debt sustainability concerns.

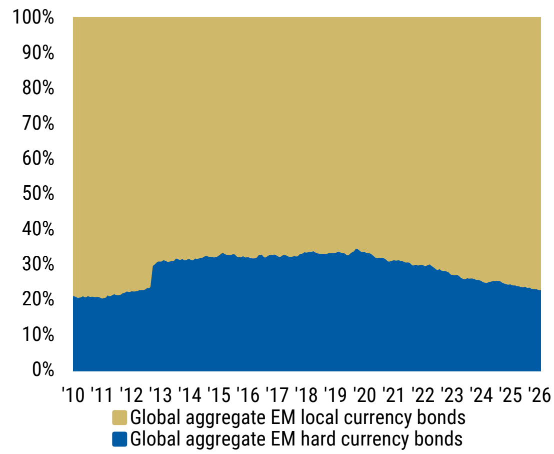
Since the pandemic, EM local-currency bonds proliferated. While issuer-related idiosyncrasies exist in EM, at the macro level, we think this development should reduce investor concerns of debt sustainability-related risk-off episodes in EM.

Exhibit 16: Bloomberg global aggregate index: EM hard and local currency bonds outstanding



Source: Morgan Stanley Research, Bloomberg

Exhibit 17: Bloomberg global aggregate index: EM hard and local currency bonds outstanding shares



Source: Morgan Stanley Research, Bloomberg

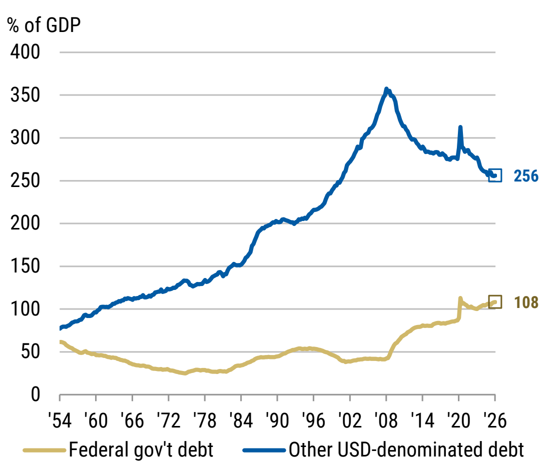
Government debt competes with other investments

Government debt does not exist in a vacuum, but rather in a large and complex world of alternative investments. Government "earnings," mainly taxes, occur with a higher probability than corporate earnings. As they say, nothing is certain except death and taxes.

Even within a country, debt sustainability concerns relating to government debt miss the forest for the trees. How many investors expressed concern about the build-up in non-governmental debt-to-GDP before the GFC?

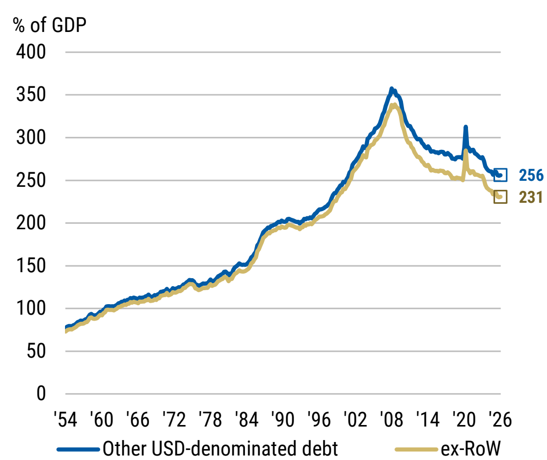
Between 1997 and 2007, USD-denominated debt outside the US federal government increased from 200% to 300% of GDP, and most investors shrugged (see Exhibit 18 and Exhibit 19). Investors have a higher probability of getting paid back on federal government debt than on non-federal debt, which remains more than twice as large relative to GDP. Tax revenues arrive with a higher probability than corporate revenues.

Exhibit 18: US federal government debt and all other USD-denominated debt as % of GDP



Source: Morgan Stanley Research, Federal Reserve

Exhibit 19: Other USD-denominated debt including and excluding rest of world debt

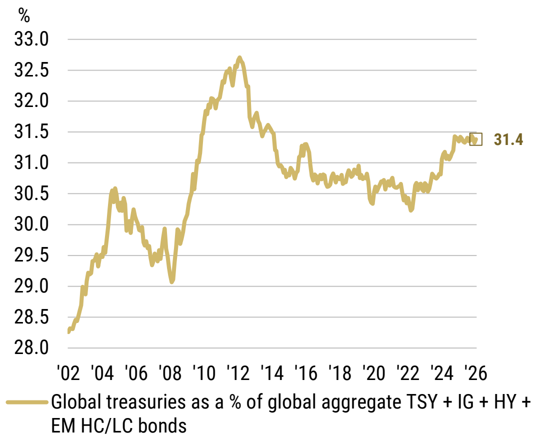


Source: Morgan Stanley Research, Federal Reserve

Proper research also needs to frame sovereign debt sustainability concerns in the appropriate context: the realm of alternative investments offering different types of risk. Exhibit 20 shows government bonds outstanding in proportion to all debt included in the global aggregate index, and Exhibit 21 adds global equities outstanding.

- Global government bonds increased as a percentage of all globally indexed debt since the pandemic, but remain below their post-GFC peak.
- When including equities, global government bonds decreased as a percentage of these alternative investments.

Exhibit 20: Global government bonds outstanding as a % of global TSY, IG, HY, EM HC/LC bonds



Source: Morgan Stanley Research, Bloomberg

Exhibit 21: Global government bonds outstanding as a % of global TSY, IG, HY, EM HC/LC bonds, and equities

Source: Morgan Stanley Research, Bloomberg

Dealer positioning shows limited crowding out

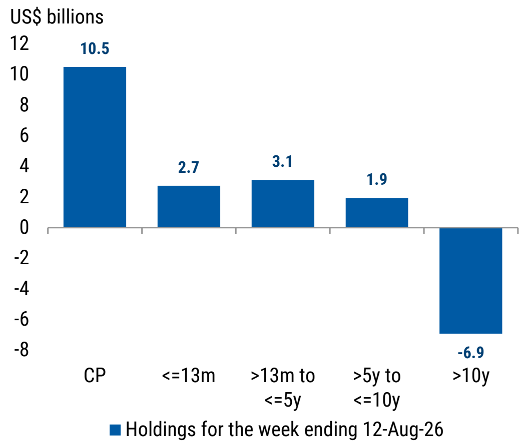
The most current version of the alternative-investment argument runs through corporate issuance. Total corporate debt issuance, most of which carries an investment grade (IG) rating, reached $1.6 trillion through July, more than 25% above the same period last year, according to SIFMA, and debt raised to fund artificial intelligence (AI) infrastructure supplied a large share of the increase.

One interpretation holds that the deluge of corporate debt has crimped demand for long-term US government debt and sent yields higher than the US administration would like. Treasury Secretary Bessent described the dynamic more narrowly. He pointed to a thinly traded market in August and to corporate issuance that influenced prices in a period of lower primary dealer transaction volumes.

In preparation for the September calendar, and after having gotten burned on prior investment, investors may have sold bonds to dealers, and dealers may have sold duration in a thin market (see Exhibit 22).

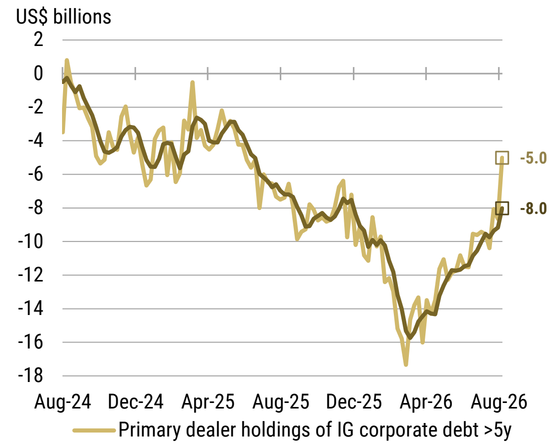
Primary dealers hold a net short position in long-end IG bonds, according to the New York Fed, but that masks primary dealer balance sheets absorbing a flood of unwanted corporate bond duration (see Exhibit 23).

Exhibit 22: Primary dealer balance sheet holdings of CP and IG corporate debt by maturity



Source: Morgan Stanley Research, FRB New York

Exhibit 23: Primary dealer balance sheet holdings of IG corporate debt >5-years to maturity



Source: Morgan Stanley Research, FRB New York

The price sensitivity of the issuers deserves more attention than it receives. Some accounts describe AI-related corporate issuance as yield agnostic, on the view that expected returns run so high that the coupon does not affect the decision to borrow.

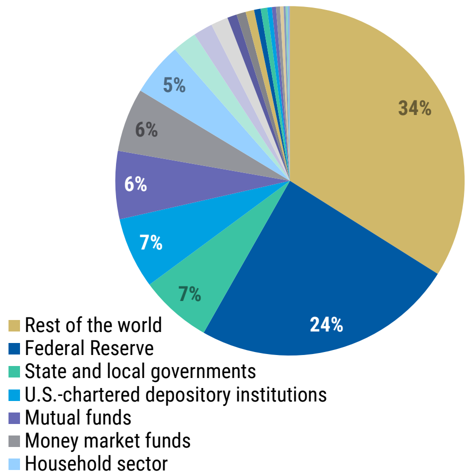
We doubt that description survives a wide enough move in yields. A borrower targeting a 25% return on invested capital and a profit margin of 15% to 20% should begin to slow issuance well before corporate yields reach the upper end of a 5% to 10% range. Investors who want to test the yield-agnostic claim should watch whether gross IG issuance from AI-related borrowers slows as yields rise, rather than whether the aggregate calendar keeps growing.

Foreign ownership shares changed little since 2022

The amount of debt, the Federal Reserve's retreat from buying it, and the assumed waning in foreign investors' appetite have left some analysts and investors wondering who will buy it all. Whether the buyers are commercial banks, asset managers, or US households, someone will buy the Treasury securities issued by the government – or else the government wouldn't keep spending, which would reduce aggregate demand and bond yields.

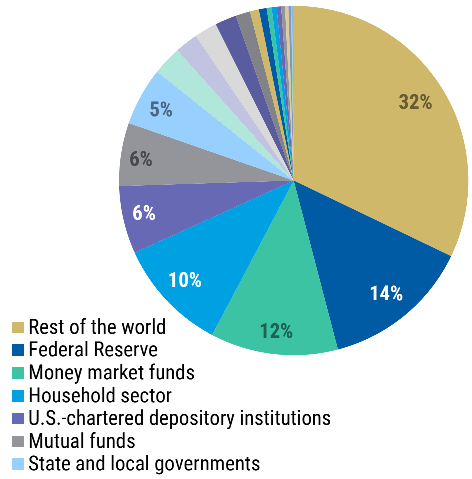
The composition of foreign holders has changed less than many investors seem to believe. Exhibit 24 and Exhibit 25 show the holders of US Treasuries in 1Q26 (the latest quarter of data on hand) and 3Q22 (when the Fed's balance sheet began to shrink). The rest of the world owns 32% of the Treasury market as of March 2026 – only down 2pp since 3Q22 vs. the Fed's holdings down 10pp over the same period.

Exhibit 24: Top 10 holders, in order, of US Treasuries (bills, notes, bonds): 3Q22



Source: Morgan Stanley Research, Federal Reserve

Exhibit 25: Top 10 holders, in order, of US Treasuries (bills, notes, bonds): 1Q26

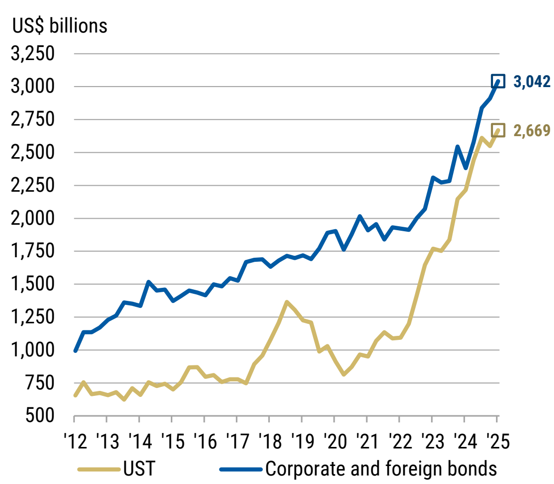


Source: Morgan Stanley Research, Federal Reserve

Of that 12pp fewer US Treasuries that the Fed and foreigners own, money market funds and the household sector, which includes hedge funds, took up nearly all of it. Since 3Q22, 10y Treasury yields have risen between 50-75bp – about the same amount as the target fed funds rate rose (3.25% → 3.75%).

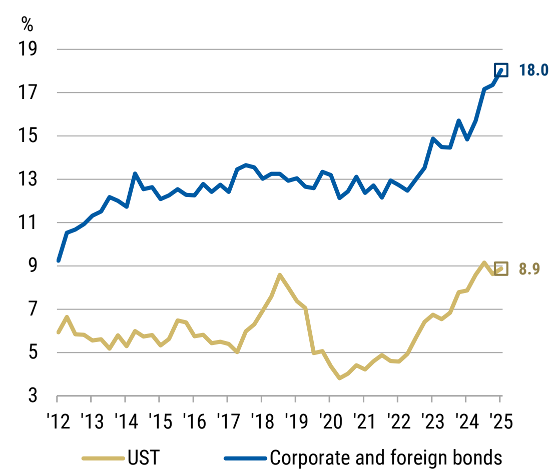
Hedge funds have grown as a holder of Treasuries, and they have grown faster as a holder of corporate and foreign bonds (see Exhibit 26 and Exhibit 27). Any argument that hedge fund demand for corporate paper diverts capital away from Treasuries needs to account for the fact that the same investor group raised its share of both markets.

Exhibit 26: Hedge fund holdings of USTs and corporate and foreign bonds



Source: Morgan Stanley Research, Federal Reserve

Exhibit 27: Hedge fund holdings of USTs and corporate and foreign bonds as % of outstanding

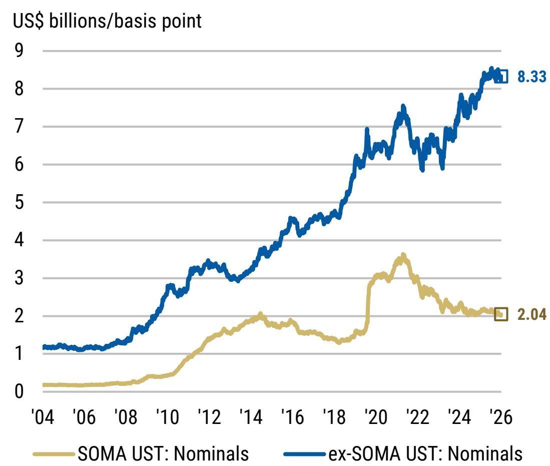


Source: Morgan Stanley Research, Federal Reserve

Descriptions of foreign investors as price-insensitive holders exiting the scene – which we showed happened to a significantly smaller degree than the popular narrative suggests – deserve scrutiny. Foreign investors, both official institutions and private sector investors, respond to price. We know this from firsthand experience meeting central bank reserve managers across Europe and Asia.

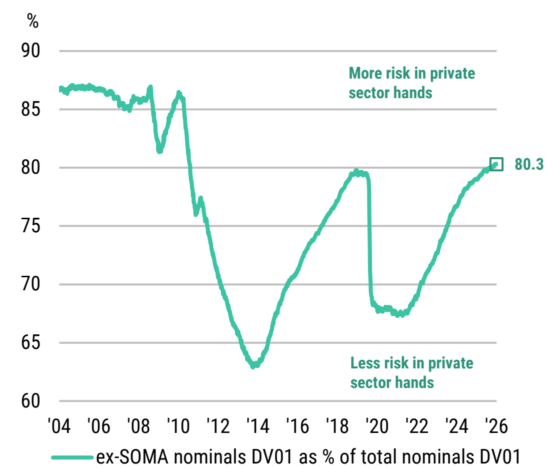
Of course, with the Fed stepping away, more of the duration risk in the Treasury market now sits with private investors. Exhibit 28 and Exhibit 29 show the DV01 of all nominal coupon Treasuries held in and outside of SOMA, with the ex-SOMA share at roughly 80% of the total. That share has followed the same path since 2022, so it adds little new information for investors. In addition, it still sits lower than it was before the GFC.

Exhibit 28: DV01 of all nominal coupon US Treasuries held in and outside of SOMA



Source: Morgan Stanley Research, US Treasury, Federal Reserve, Bloomberg

Exhibit 29: ex-SOMA nominal coupon UST DV01 as a % of total nominal coupon UST DV01



Source: Morgan Stanley Research, US Treasury, Federal Reserve, Bloomberg

If not the Fed, then who will buy the forthcoming supply?

This common question reveals a misunderstanding of how investors (and the markets in which they move price) future supply. The question seems understandable, given the amount of bonds that investors need to purchase. Given our goal of helping investors add alpha to their portfolio, we do not see value in estimating who will buy these bonds because it would not help us predict the direction of bond prices/yields.

Even if we could provide an accurate account of which investors would buy, we would also have to forecast when those investors would buy and at what price. Knowing who will buy the supply has little meaning in the absence of knowing the timing and price at which those purchases occur.

What determines timing and price at which investors make the decision whether to buy, hold, or sell? More often than not, the answer comes back to the macro environment and its impact on central bank decision making.

In the absence of knowing who will buy, when, and at what price, how should investors think about supply and its impact on price? We discuss the impact of supply on price in Between a Rock and a Hard Place. In short, sovereign supply moves market prices the most when expectations about it form or change from previously held expectations.

Daily turnover lets marginal buyers set prices

Once we agree that the US government has the ability to pay back its debt, in every circumstance, then we can debate whether it has the willingness to do so. Provided we agree it has the willingness, then the money necessary to finance future deficits comes from the government's decision to increase spending or accept less tax revenue.

In the end, a federal government deficit amounts to nothing more than an accounting entry on the government balance sheet – a balance sheet that differs materially from that of the corporate and household sector, given the federal government's ability to pay back its obligations.

Government deficits create either public sector demand, via spending, or private sector demand, via tax cuts and financial assets, and the money the government spends, or the money that is spent and saved after a tax cut, eventually finds its way into US Treasuries.

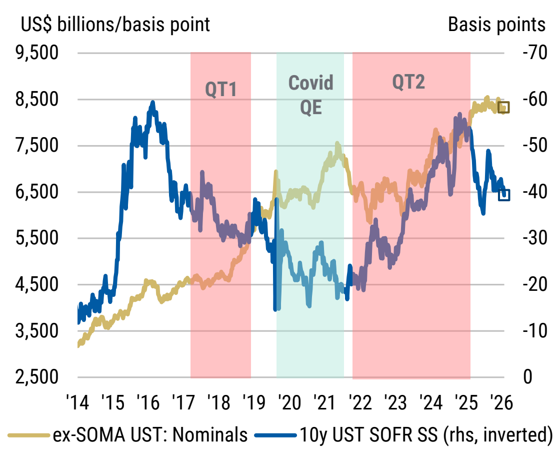
The relevant question for macro investors, however, concerns the price at which Treasury securities clear rather than the identity of the buyer. Swap spreads offer one way to examine that price.

For relative value investors, like those dealing in swap spreads, the "who" matters, but macro investors should rest assured someone will buy, even if they don't know who.

By comparing cash Treasury yields with matched-maturity SOFR swap rates, they provide a measure of Treasury relative value in which changes in Treasury supply should appear more directly than in outright yields.

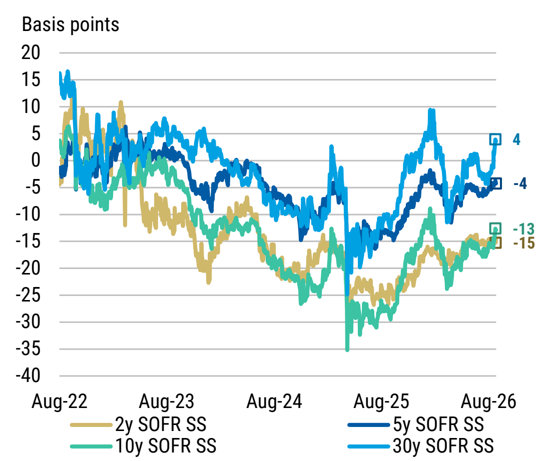
Put differently, spreads between cash Treasuries and their SOFR swap rate equivalents offer another way to distill concern over the *amount* the amount of US public debt, from the *price* at which investors will hold it.

Exhibit 30: Ex-SOMA nominal coupon UST DV01 and 10y UST SOFR SS, under different Fed balance sheet regimes



Source: Morgan Stanley Research, Bloomberg

Exhibit 31: Change in UST SOFR swap spreads since US public debt reached $31tr on October 1, 2022



Source: Morgan Stanley Research, Bloomberg

Higher deficits generally require greater Treasury issuance. All else equal, more Treasury supply should cheapen cash Treasuries relative to swaps, and tighten swap spreads.

The historical relationship between the amount of Treasury duration held outside SOMA and 10y swap spreads shows that effect – but also its limits (see Exhibit 30).

Even where Treasury supply should matter most directly – in the relative price of Treasuries versus swaps – the relationship is not mechanical.

A near-record amount of nominal Treasury DV01 sits in private sector hands, yet 10y swap spreads have widened more than 15bp since QT2 ended in December 2025.

Since US public debt surpassed $31tr in October 2022, the subsequent $9tr increase coincided with only 13bp of tightening in 10y swap spreads but 4bp of *widening* in 30y spreads (see Exhibit 31). The stock of debt rose; its relative price did not move uniformly.

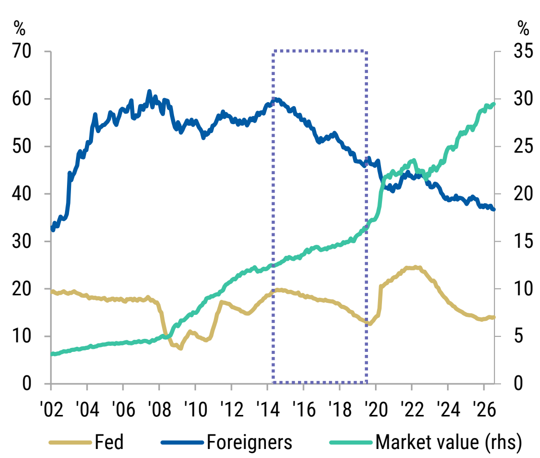
Even where Treasury supply should appear most directly, Fed balance-sheet policy, regulation, investor demand, financing conditions and dealer intermediation capacity also influence the price at which Treasuries clear.

That distinction brings us back to who buys the debt? Knowing the identity of the largest holders tells us little about the price established by the marginal buyer or seller.

History supplies a useful check. From 4Q14 through 3Q19, both the Fed and foreign investors reduced Treasury holdings in market value terms, despite the increase in the market value of all Treasuries (see Exhibit 32). Treasury yields trended lower during this period, with 10-year yields reaching multi-decade lows at 1.32% in July 2016.

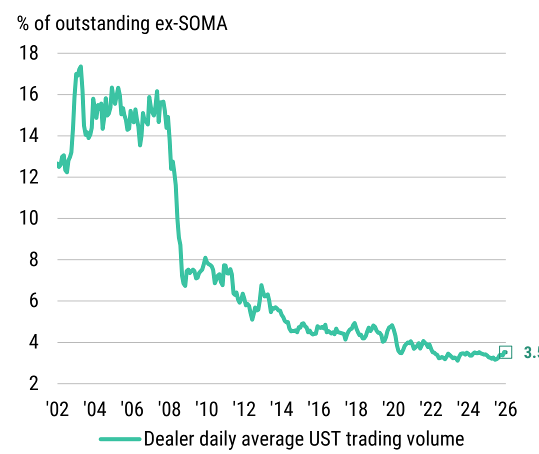
That period, among others, suggests that even large investor types don't dictate the overall level of UST yields. The marginal buyer or seller, which could end up being any number of investor types, moves prices, not the largest. At least 3.5% of outstanding USTs change hands every day, an open invitation for many investors, including those who use leverage, to move prices (see Exhibit 33).

Exhibit 32: Fed SOMA holdings as a % of outstanding vs foreign holdings as a % of outstanding ex-SOMA



Source: Morgan Stanley Research, FRB Dallas, US Treasury, Bloomberg

Exhibit 33: Primary dealer average daily UST trading volume as a % of UST debt outstanding ex-SOMA



Source: Morgan Stanley Research, FRB New York, US Treasury, Bloomberg

Fiscal variables do not map stably to long rates

In the end, investors, not models, judge debt sustainability. Academic approaches use models with variables such as real interest rates (r) and economic growth (g), while investors must judge for themselves the acceptable yield to take on sustainability risks.

Research that addresses debt sustainability usually leaves open the question of how investors judge sustainability themselves and how prices differ for sustainable versus unsustainable debt loads. Markets price many risks concurrently, which makes judging the proportion of a government bond yield related to debt sustainability concerns challenging, and the component of a yield thought to house sustainability concerns, the term premium, houses many other investor concerns as well.

Academic research has tried to link debt and deficits to movements in government bond yields. Researchers at the International Monetary Fund, Furceri et al (2025), replicated the approach used by Laubach (2009) and previously Laubach (2003), and their conclusions mirrored those of Laubach: the link between long-term rates and fiscal variables does not remain stable over time.

- The inconsistency in Laubach's results likely comes from the context in which markets incorporate expectations for supply and how central banks respond. A counter-cyclical stimulus, such as the CARES Act, or a pro-cyclical stimulus, such as the Tax Cuts and Jobs Act passed in 2017, provide dramatically different contexts, and were accompanied by dramatically different central bank actions.

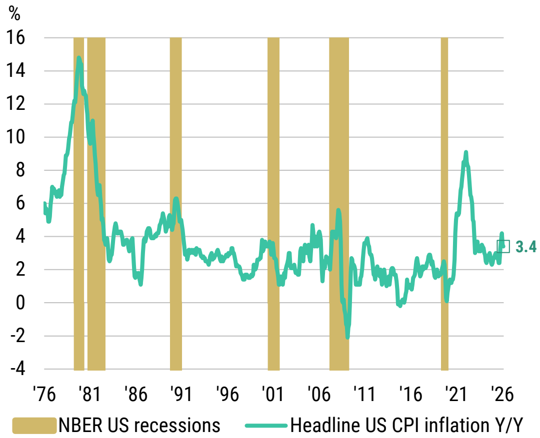
Regression analysis of the type used in the IMF and Laubach papers also suffers from the absence of proper data. A proper analysis of the effect of debt accumulation on government bond yields requires a time series of investor expectations relating to government bond supply, not a semi-annual projection of 10-year forward federal government budget deficits from the CBO.

Large deficits over short periods can produce inflation

Of the four variables that matter, the speed at which the debt accumulates carries the clearest historical link to markets. We know how too large a deficit in too short a period of time can lead to runaway consumer price inflation. The post-pandemic period is "Exhibit A" (see Exhibit 34).

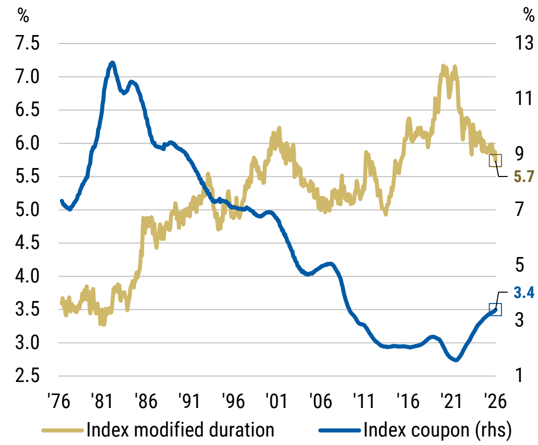
The risk characteristics of the debt matter as well. We know how the average duration of the debt and the average coupon it pays can affect investor returns for the better, and worse. Today, Treasuries offer a below-average coupon at an above average duration (see Exhibit 35).

Exhibit 34: US headline CPI inflation Y/Y vs NBER US recessions



Source: Morgan Stanley Research, BLS, Bloomberg

Exhibit 35: Bloomberg US Treasury index modified duration and average coupon



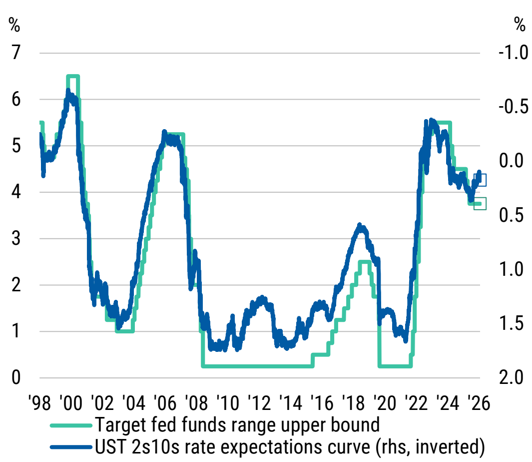
Source: Morgan Stanley Research, Bloomberg

The Fed drives the level of Treasury yields

So what determines the level of Treasury yields over time? The Fed and its monetary policies play the leading role, while fiscal policy and overseas monetary policies play supporting roles.

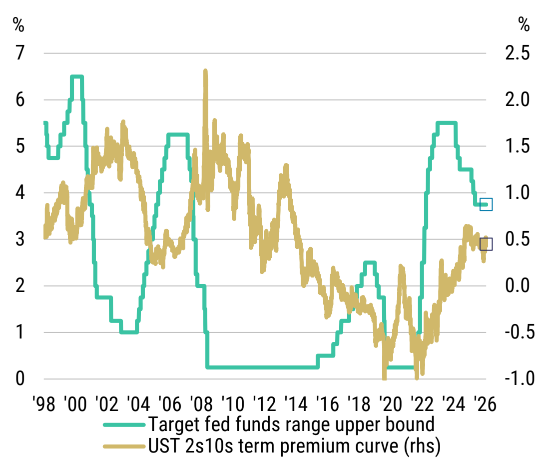
Decomposing Treasury yields into rate expectations and term premiums makes this clear (see Exhibit 36 and Exhibit 37). The Fed's interest rate policy, and forward guidance around it, drives rate expectations, and rate expectations have been the largest driver of 10y yields historically. 10y rate expectations moved in a 500 basis point (bp) range prior to the advent of quantitative easing (QE), whereas 10y term premiums moved in a 300bp range.

Exhibit 36: ACM 10y rate expectations and target fed funds rate range midpoint



Source: Morgan Stanley Research, Federal Reserve, FRB New York

Exhibit 37: ACM 10y term premium and target fed funds rate range midpoint



Source: Morgan Stanley Research, Federal Reserve, FRB New York

Interest rate policy also drove term premiums prior to the advent of QE. As the Fed raised short-term rates, term premiums fell, presumably because investors believed that tighter monetary policy would weigh on growth and inflation, a belief that was reflected in lower risk premiums.

Since late 2008, however, the Fed's balance sheet policy and forward guidance around it have played a larger role in term premium movements. In addition, the interest rate and balance sheet policies of other central banks have played important roles.

In conclusion, we suggest macro investors pay more attention to central bank policies and reaction functions, and the economic data that feed into them, than the overall amount of government debt investors will need to purchase, or which investors will do the buying.

With apologies to the famous line from the movie "Field of Dreams", if you sell it at the wrong price, a buyer will come.

Global Macro Strategy Team

MORGAN STANLEY & CO. LLC	**Matthew Hornbach, CMT** Matthew.Hornbach@morganstanley.com	Global Head of Macro Strategy	+1 212 761-1837
	Martin Tobias, CFA, CMT	US Rates Strategist	+1 212 761-6076
	Shaun Zhou	US Rates Strategist	+1 212 761-3348
	Aryaman Singh	US Rates Strategist	+1 212 761-1993
	Eli Carter	US Rates Strategist	+1 212 761-4703
MORGAN STANLEY & CO. LLC	**Andrew Watrous**	G10 FX Strategist	+1 212 761-5287
	Molly Nickolin	G10 FX Strategist	+1 212 761-3592
	Simon Waever Simon.Waever@morganstanley.com	Head of EM Sovereign Credit and Latin America Fixed Income Strategy	+1 212 296-8101
	Ioana Zamfir	Latin America Macro Strategist	+1 212 761-4012
	Emma Cerda	Latin America Sovereign Credit	+1 212 761-2344
MORGAN STANLEY & CO. INTERNATIONAL PLC	**James K. Lord** James.Lord@morganstanley.com	Global Head of FXEM Strategy	+44 20 7677-3254
	Gianluca Salford Luca.Salford@morganstanley.com	Head of European Rates Strategy	+44 20 7677-1337
	Fabio Bassanin, CFA	UK Rates Strategist	+44 20 7425-1869
	Maria Chiara Russo	Euro Area Rates Strategist	+44 20 7677-3499
	Jasper Knyphausen	Euro Area Rates Strategist	+44 20 7425-2185
	David S. Adams, CFA David.S.Adams@morganstanley.com	Head of G10 FX Strategy	+44 20 7425-3518
	Alexander Scholefield	G10 FX Strategist	+44 20 7425-3518
	Neville Mandimika	CEEMEA Sovereign Credit Strategist	+44 20 7425-2509
	Arnav Gupta	CEEMEA Rates Strategist	+44 20 7677-0382
MORGAN STANLEY ASIA LIMITED+	**Gek Teng Khoo**	AXJ Macro Strategist	+852 3963-0303
MORGAN STANLEY MUFG SECURITIES CO., LTD.	**Koichi Sugisaki** Koichi.Sugisaki@morganstanleymufg.com	Head of Japan Macro Strategy	+81 3 6836-8428
	Hiromu Uezato	Japan Macro Strategist	+81 3 6836-8431

Disclosure Section

The information and opinions in Morgan Stanley Research were prepared by Morgan Stanley & Co. LLC, and/or Morgan Stanley C.T.V.M. S.A., and/or Morgan Stanley Mexico, Casa de Bolsa, S.A. de C.V., and/or Morgan Stanley Canada Limited. As used in this disclosure section, "Morgan Stanley" includes Morgan Stanley & Co. LLC, Morgan Stanley C.T.V.M. S.A., Morgan Stanley Mexico, Casa de Bolsa, S.A. de C.V., Morgan Stanley Canada Limited and their affiliates as necessary.

For important disclosures, stock price charts and equity rating histories regarding companies that are the subject of this report, please see the Morgan Stanley Research Disclosure Website at www.morganstanley.com/eqr/disclosures/webapp/generalresearch, or contact your investment representative or Morgan Stanley Research at 1585 Broadway, (Attention: Research Management), New York, NY, 10036 USA.

For valuation methodology and risks associated with any recommendation, rating or price target referenced in this research report, please contact the Client Support Team as follows: US/Canada +1 800 303-2495; Hong Kong +852 2848-5999; Latin America +1 718 754-5444 (U.S.); London +44 (0)20-7425-8169; Singapore +65 6834-6860; Sydney +61 (0)2-9770-1505; Tokyo +81 (0)3-6836-9000. Alternatively you may contact your investment representative or Morgan Stanley Research at 1585 Broadway, (Attention: Research Management), New York, NY 10036 USA.

Analyst Certification

The following analysts hereby certify that their views about the companies and their securities discussed in this report are accurately expressed and that they have not received and will not receive direct or indirect compensation in exchange for expressing specific recommendations or views in this report: Eli P Carter; Matthew Hornbach; Aryaman Singh; Martin W Tobias, CFA; Shaun Zhou.

.

Global Research Conflict Management Policy

Morgan Stanley Research has been published in accordance with our conflict management policy, which is available at www.morganstanley.com/institutional/research/conflictpolicies. A Portuguese version of the policy can be found at www.morganstanley.com.br

Important Regulatory Disclosures on Subject Companies

In the next 3 months, Morgan Stanley expects to receive or intends to seek compensation for investment banking services from United States of America.

Within the last 12 months, Morgan Stanley has provided or is providing investment banking services to, or has an investment banking client relationship with, the following company: United States of America.

The equity research analysts or strategists principally responsible for the preparation of Morgan Stanley Research have received compensation based upon various factors, including quality of research, investor client feedback, stock picking, competitive factors, firm revenues and overall investment banking revenues. Equity Research analysts' or strategists' compensation is not linked to investment banking or capital markets transactions performed by Morgan Stanley or the profitability or revenues of particular trading desks.

Morgan Stanley and its affiliates do business that relates to companies/instruments covered in Morgan Stanley Research, including market making, providing liquidity, fund management, commercial banking, extension of credit, investment services and investment banking. Morgan Stanley sells to and buys from customers the securities/instruments of companies covered in Morgan Stanley Research on a principal basis. Morgan Stanley may have a position in the debt of the Company or instruments discussed in this report. Morgan Stanley trades or may trade as principal in the debt securities (or in related derivatives) that are the subject of the debt research report.

Certain disclosures listed above are also for compliance with applicable regulations in non-US jurisdictions.

STOCK RATINGS

Morgan Stanley uses a relative rating system using terms such as Overweight, Equal-weight, Not-Rated or Underweight (see definitions below). Morgan Stanley does not assign ratings of Buy, Hold or Sell to the stocks we cover. Overweight, Equal-weight, Not-Rated and Underweight are not the equivalent of buy, hold and sell. Investors should carefully read the definitions of all ratings used in Morgan Stanley Research. In addition, since Morgan Stanley Research contains more complete information concerning the analyst's views, investors should carefully read Morgan Stanley Research, in its entirety, and not infer the contents from the rating alone. In any case, ratings (or research) should not be used or relied upon as investment advice. An investor's decision to buy or sell a stock should depend on individual circumstances (such as the investor's existing holdings) and other considerations.

Global Stock Ratings Distribution

(as of July 31, 2026)

The Stock Ratings described below apply to Morgan Stanley's Fundamental Equity Research and do not apply to Debt Research produced by the Firm.

For disclosure purposes only (in accordance with FINRA requirements), we include the category headings of Buy, Hold, and Sell alongside our ratings of Overweight, Equal-weight, Not-Rated and Underweight. Morgan Stanley does not assign ratings of Buy, Hold or Sell to the stocks we cover. Overweight, Equal-weight, Not-Rated and Underweight are not the equivalent of buy, hold, and sell but represent recommended relative weightings (see definitions below). To satisfy regulatory requirements, we correspond Overweight, our most positive stock rating, with a buy recommendation; we correspond Equal-weight and Not-Rated to hold and Underweight to sell recommendations, respectively.

	Coverage Universe		Investment Banking Clients (IBC)			Other Material Investment Services Clients (MISC)	
Stock Rating Category	Count	% of Total	Count	% of Total IBC	% of Rating Category	Count	% of Total Other MISC
Overweight/Buy	1529	42%	442	47%	29%	750	43%
Equal-weight/Hold	1582	43%	396	42%	25%	773	44%
Not-Rated/Hold	3	0%	1	0%	33%	1	0%
Underweight/Sell	560	15%	97	10%	17%	217	12%
Total	3,674		936			1741	

Data include common stock and ADRs currently assigned ratings. Investment Banking Clients are companies from whom Morgan Stanley received investment banking compensation in the last 12 months. Due to rounding off of decimals, the percentages provided in the "% of total" column may not add up to exactly 100 percent.

Analyst Stock Ratings

Overweight (O). The stock's total return is expected to exceed the average total return of the analyst's industry (or industry team's) coverage universe, on a risk-adjusted basis, over the next 12-18 months.

Equal-weight (E). The stock's total return is expected to be in line with the average total return of the analyst's industry (or industry team's) coverage universe, on a risk-adjusted basis, over the next 12-18 months.

Not-Rated (NR). Currently the analyst does not have adequate conviction about the stock's total return relative to the average total return of the analyst's industry (or industry team's) coverage universe, on a risk-adjusted basis, over the next 12-18 months.

Underweight (U). The stock's total return is expected to be below the average total return of the analyst's industry (or industry team's) coverage universe, on a risk-adjusted basis, over the next 12-18 months.

Unless otherwise specified, the time frame for price targets included in Morgan Stanley Research is 12 to 18 months.

Analyst Industry Views

Attractive (A): The analyst expects the performance of his or her industry coverage universe over the next 12-18 months to be attractive vs. the relevant broad market benchmark, as indicated below.

In-Line (I): The analyst expects the performance of his or her industry coverage universe over the next 12-18 months to be in line with the relevant broad market benchmark, as indicated below.

Cautious (C): The analyst views the performance of his or her industry coverage universe over the next 12-18 months with caution vs. the relevant broad market benchmark, as indicated below.

Benchmarks for each region are as follows: North America - S&P 500; Latin America - relevant MSCI country index or MSCI Latin America Index; Europe - MSCI Europe; Japan - TOPIX; Asia - relevant MSCI country index or MSCI sub-regional index or MSCI AC Asia Pacific ex Japan Index.

Important Disclosures for Morgan Stanley Smith Barney LLC Customers

Important disclosures regarding any material conflict of interest that can reasonably be expected to have influenced Morgan Stanley Smith Barney LLC's choice of a third-party research provider or the subject company of a third-party research report, are available on the Morgan Stanley Wealth Management disclosure website at www.morganstanley.com/online/researchdisclosures.

For Morgan Stanley specific disclosures, you may refer to https://www.morganstanley.com/eqr/disclosures/webapp/generalresearch.

Each Morgan Stanley research report is reviewed and approved on behalf of Morgan Stanley Smith Barney LLC. This review and approval is conducted by the same person who reviews the research report on behalf of Morgan Stanley. This could create a conflict of interest.

Other Important Disclosures

Morgan Stanley Research policy is to update research reports as and when the Research Analyst and Research Management deem appropriate, based on developments with the issuer, the sector, or the market that may have a material impact on the research views or opinions stated therein. In addition, certain Research publications are intended to be updated on a regular periodic basis (weekly/monthly/quarterly/annual) and will ordinarily be updated with that frequency, unless the Research Analyst and Research Management determine that a different publication schedule is appropriate based on current conditions.

Morgan Stanley is not acting as a municipal advisor and the opinions or views contained herein are not intended to be, and do not constitute, advice within the meaning of Section 975 of the Dodd-Frank Wall Street Reform and Consumer Protection Act.

Morgan Stanley produces an equity research product called a "Tactical Idea." Views contained in a "Tactical Idea" on a particular stock may be contrary to the recommendations or views expressed in research on the same stock. This may be the result of differing time horizons, methodologies, market events, or other factors. For all research available on a particular stock, please contact your sales representative or go to Matrix at http://www.morganstanley.com/matrix.

Morgan Stanley Research is provided to our clients through our proprietary research portal on Matrix and also distributed electronically by Morgan Stanley to clients. Certain, but not all, Morgan Stanley Research products are also made available to clients through third-party vendors or redistributed to clients through alternate electronic means as a convenience. For access to all available Morgan Stanley Research, please contact your sales representative or go to Matrix at http://www.morganstanley.com/matrix.

Any access and/or use of Morgan Stanley Research is subject to Morgan Stanley's Terms of Use (http://www.morganstanley.com/terms.html). By accessing and/or using Morgan Stanley Research, you are indicating that you have read and agree to be bound by our Terms of Use (http://www.morganstanley.com/terms.html). In addition you consent to Morgan Stanley processing your personal data and using cookies in accordance with our Privacy Policy and our Global Cookies Policy (http://www.morganstanley.com/privacy_pledge.html), including for the purposes of setting your preferences and to collect readership data so that we can deliver better and more personalized service and products to you. To find out more information about how Morgan Stanley processes personal data, how we use cookies and how to reject cookies see our Privacy Policy and our Global Cookies Policy (http://www.morganstanley.com/privacy_pledge.html). Please use the provided link to review the Terms and Conditions and Most Important Terms and Conditions for Morgan Stanley India Company Private Limited (https://www.morganstanley.com/assets/pdfs/about-us-global-offices/india/Terms_and_conditions.pdf) and the following link to review the audit report (https://ny.matrix.ms.com/eqr/research/webapp/researchdocs/MSICPL_Morgan_Stanley_Research_Audit_Report.pdf).

If you do not agree to our Terms of Use and/or if you do not wish to provide your consent to Morgan Stanley processing your personal data or using cookies please do not access our research.

Morgan Stanley Research does not provide individually tailored investment advice. Morgan Stanley Research has been prepared without regard to the circumstances and objectives of those who receive it. Morgan Stanley recommends that investors independently evaluate particular investments and strategies, and encourages investors to seek the advice of a financial adviser. The appropriateness of an investment or strategy will depend on an investor's circumstances and objectives. The securities, instruments, or strategies discussed in Morgan Stanley Research may not be suitable for all investors, and certain investors may not be eligible to purchase or participate in some or all of them. Morgan Stanley Research is not an offer to buy or sell or the solicitation of an offer to buy or sell any security/instrument or to participate in any particular trading strategy. The value of and income from your investments may vary because of changes in interest rates, foreign exchange rates, default rates, prepayment rates, securities/instruments prices, market indexes, operational or financial conditions of companies or other factors. There may be time limitations on the exercise of options or other rights in securities/instruments transactions. Past performance is not necessarily a guide to future performance. Estimates of future performance are based on assumptions that may not be realized. If provided, and unless otherwise stated, the closing price on the cover page is that of the primary exchange for the subject company's securities/instruments.

The fixed income research analysts, strategists or economists principally responsible for the preparation of Morgan Stanley Research have received compensation based upon various factors, including quality, accuracy and value of research, firm profitability or revenues (which include fixed income trading and capital markets profitability or revenues), client feedback and competitive factors. Fixed Income Research analysts', strategists' or economists' compensation is not linked to investment banking or capital markets transactions performed by Morgan Stanley or the profitability or revenues of particular trading desks.

The "Important Regulatory Disclosures on Subject Companies" section in Morgan Stanley Research lists all companies mentioned where Morgan Stanley owns 1% or more of a class of common equity securities of the companies. For all other companies mentioned in Morgan Stanley Research, Morgan Stanley may have an investment of less than 1% in securities/instruments or derivatives of securities/instruments of companies and may trade them in ways different from those discussed in Morgan Stanley Research. Employees of Morgan Stanley not involved in the preparation of Morgan Stanley Research may have investments in securities/instruments or derivatives of securities/instruments of companies mentioned and may trade them in ways different from those discussed in Morgan Stanley Research. Derivatives may be issued by Morgan Stanley or associated persons.

With the exception of information regarding Morgan Stanley, Morgan Stanley Research is based on public information. Morgan Stanley makes every effort to use reliable, comprehensive information, but we make no representation that it is accurate or complete. We have no obligation to tell you when opinions or information in Morgan Stanley Research change apart from when we intend to discontinue equity research coverage of a subject company. Facts and views presented in Morgan Stanley Research have not been reviewed by, and may not reflect information known to, professionals in other Morgan Stanley business areas, including investment banking personnel.

Morgan Stanley Research personnel may participate in company events such as site visits and are generally prohibited from accepting payment by the company of associated expenses unless pre-approved by authorized members of Research management.

Morgan Stanley may make investment decisions that are inconsistent with the recommendations or views in this report.

To our readers based in Taiwan or trading in Taiwan securities/instruments: Information on securities/instruments that trade in Taiwan is distributed by Morgan Stanley Taiwan Limited ("MSTL"). Such information is for your reference only. The reader should independently evaluate the investment risks and is solely responsible for their investment decisions. Morgan Stanley Research may not be distributed to the public media or quoted or used by the public media without the express written consent of Morgan Stanley. Any non-customer reader within the scope of Article 7-1 of the Taiwan Stock Exchange Recommendation Regulations accessing and/or receiving Morgan Stanley Research is not permitted to provide Morgan Stanley Research to any third party (including but not limited to related parties, affiliated companies and any other third parties) or engage in any activities regarding Morgan Stanley Research which may create or give the appearance of creating a conflict of interest. Information on securities/instruments that do not trade in Taiwan is for informational purposes only and is not to be construed as a recommendation or a solicitation to trade in such securities/instruments. MSTL may not execute transactions for clients in these securities/instruments.

Morgan Stanley is not incorporated under PRC law and the research in relation to this report is conducted outside the PRC. Morgan Stanley Research does not constitute an offer to sell or the solicitation of an offer to buy any securities in the PRC. PRC investors shall have the relevant qualifications to invest in such securities and shall be responsible for obtaining all relevant approvals, licenses, verifications and/or registrations from the relevant governmental authorities themselves. Neither this report nor any part of it is intended as, or shall constitute, provision of any consultancy or advisory service of securities investment as defined under PRC law. Such information is provided for your reference only.

Morgan Stanley Research is disseminated in Brazil by Morgan Stanley C.T.V.M. S.A. located at Av. Brigadeiro Faria Lima, 3600, 6th floor, São Paulo - SP, Brazil; and is regulated by the Comissão de Valores Mobiliários; in Mexico by Morgan Stanley México, Casa de Bolsa, S.A. de C.V which is regulated by Comision Nacional Bancaria y de Valores. Paseo de los Tamarindos 90, Torre 1, Col. Bosques de las Lomas Floor 29, 05120 Mexico City; in Japan by Morgan Stanley MUFG Securities Co., Ltd. and, for Commodities related research reports only, Morgan Stanley Capital Group Japan Co., Ltd; in Hong Kong by Morgan Stanley Asia Limited (which accepts responsibility for its contents) and by Morgan Stanley Bank Asia Limited; in Singapore by Morgan Stanley Asia (Singapore) Pte. (Registration number 199206298Z) and/or Morgan Stanley Asia (Singapore) Securities Pte Ltd (Registration number 200008434H), regulated by the Monetary Authority of Singapore (which accepts legal responsibility for its contents and should be contacted with respect to any matters arising from, or in connection with, Morgan Stanley Research) and by Morgan Stanley Bank Asia Limited, Singapore Branch (Registration number T14FC0118J); in Australia to "wholesale clients" within the meaning of the Australian Corporations Act by Morgan Stanley Australia Limited A.B.N. 67 003 734 576, holder of Australian financial services license No. 233742, which accepts responsibility for its contents; in Australia to "wholesale clients" and "retail clients" within the meaning of the Australian Corporations Act by Morgan Stanley Wealth Management Australia Pty Ltd (A.B.N. 19 009 145 555, holder of Australian financial services license No. 240813, which accepts responsibility for its contents; in Korea by Morgan Stanley & Co International plc, Seoul Branch; in India by Morgan Stanley India Company Private Limited having Corporate Identification No (CIN) U22990MH1998PTC115305, regulated by the Securities and Exchange Board of India ("SEBI") and holder of licenses as a Research Analyst (SEBI Registration No. INH000001105); Stock Broker (SEBI Stock Broker Registration No. INZ000244438), Merchant Banker (SEBI Registration No. INM000011203), and depository participant with National Securities Depository Limited (SEBI Registration No. IN-DP-NSDL-567-2021) having registered office at Altimus, Level 39 & 40, Pandurang Budhkar Marg, Worli, Mumbai 400018, India; Telephone no. +91-22-61181000; Compliance Officer Details: Mr. Tejarshi Hardas, Tel. No.: +91-22-61181000 or Email: tejarshi.hardas@morganstanley.com; Grievance officer details: Mr. Tejarshi Hardas, Tel. No.: +91-22-61181000 or Email: msic-compliance@morganstanley.com. Morgan Stanley India Company Private Limited (MSICPL) may use AI tools in providing research services. All recommendations contained herein are made by the duly qualified research analysts; in Canada by Morgan Stanley Canada Limited; in Germany and the European Economic Area where required by Morgan Stanley Europe S.E., authorised and regulated by Bundesanstalt fuer Finanzdienstleistungsaufsicht (BaFin) under the reference number 149169; in the US by Morgan Stanley & Co. LLC, which accepts responsibility for its contents. Morgan Stanley & Co. International plc, authorized by the Prudential Regulation Authority and regulated by the Financial Conduct Authority and the Prudential Regulation Authority, disseminates in the UK research that it has prepared, and research which has been prepared by any of its affiliates, only to persons who (i) are investment professionals falling within Article 19(5) of the Financial Services and Markets Act 2000 (Financial Promotion) Order 2005 (as amended, the "Order"); (ii) are persons who are high net worth entities falling within Article 49(2)(a) to (d) of the Order; or (iii) are persons to whom an invitation or inducement to engage in investment activity (within the meaning of section 21 of the Financial Services and Markets Act 2000, as amended) may otherwise lawfully be communicated or caused to be communicated. RMB Morgan Stanley Proprietary Limited is a member of the JSE Limited and A2X (Pty) Ltd. RMB Morgan Stanley Proprietary Limited is a joint venture owned equally by Morgan Stanley International Holdings Inc. and RMB Investment Advisory (Proprietary) Limited, which is wholly owned by FirstRand Limited. The information in Morgan Stanley Research is being disseminated by Morgan Stanley Saudi Arabia, regulated by the Capital Market Authority in the Kingdom of Saudi Arabia, and is directed at Sophisticated investors only.

The information in Morgan Stanley Research is being communicated by Morgan Stanley & Co. International plc (DIFC Branch), regulated by the Dubai Financial Services Authority (the DFSA) or by Morgan Stanley & Co. International plc (ADGM Branch), regulated by the Financial Services Regulatory Authority Abu Dhabi (the FSRA), and is directed at Professional Clients only, as defined by the DFSA or the FSRA, respectively. The financial products or financial services to which this research relates will only be made available to a customer who we are satisfied meets the regulatory criteria of a Professional Client. A distribution of the different MS Research ratings or recommendations, in percentage terms for Investments in each sector covered, is available upon request from your sales representative.

The information in Morgan Stanley Research is being communicated by Morgan Stanley & Co. International plc (QFC Branch), regulated by the Qatar Financial Centre Regulatory Authority (the QFCRA), and is directed at business customers and market counterparties only and is not intended for Retail Customers as defined by the QFCRA.

As required by the Capital Markets Board of Turkey, investment information, comments and recommendations stated here, are not within the scope of investment advisory activity. Investment advisory service is provided exclusively to persons based on their risk and income preferences by the authorized firms. Comments and recommendations stated here are general in nature. These opinions may not fit to your financial status, risk and return preferences. For this reason, to make an investment decision by relying solely to this information stated here may not bring about outcomes that fit your expectations.

The trademarks and service marks contained in Morgan Stanley Research are the property of their respective owners. Third-party data providers make no warranties or representations relating to the accuracy, completeness, or timeliness of the data they provide and shall not have liability for any damages relating to such data. The Global Industry Classification Standard (GICS) was developed by and is the exclusive property of MSCI and S&P.

Indicators and trackers referenced in Morgan Stanley Research may not be used as, or treated as, a benchmark under Regulation EU 2016/1011, or any other similar framework.

The issuers and/or fixed income products recommended or discussed in certain fixed income research reports may not be continuously followed. Accordingly, investors should regard those fixed income research reports as providing stand-alone analysis and should not expect continuing analysis or additional reports relating to such issuers and/or individual fixed income products.

Morgan Stanley may hold, from time to time, material financial and commercial interests regarding the company subject to the Research report.

Registration granted by SEBI and certification from the National Institute of Securities Markets (NISM) in no way guarantee performance of the intermediary or provide any assurance of returns to investors. Investment in securities market are subject to market risks. Read all the related documents carefully before investing.

The following authors are Fixed Income Research Analysts/Strategists and are not opining on or expressing recommendations on equity securities: Aryaman Singh; Shaun Zhou; Eli P Carter; Matthew Hornbach; Martin W Tobias, CFA.